Project Title : Osteojoint Powder product insert

Actual Size : 260mm x 110mm

Draft 24 Apr 2025

K: 100%

ELDON Nutrition

OSTEOJOINT POWDER FOR ORAL SOLUTION

GLUCOSAMINE 1,500MG | CHONDROITIN 1,200MG | MSM 1,000MG

Active Ingredient(s):	
Each sachet contains:	
Glucosamine Sulphate Sodium Chloride (eq. to 1,500 mg Glucosamine Sulphate) *(derived from seafood)*	1,884 mg
Chondroitin Sulphate Sodium (eq. to 1,200 mg Chondroitin Sulphate) *(source from bovine)*	1,320 mg
Methylsulfonylmethane (MSM)	1,000 mg

Presentations:
Boxes of 8.5 g x 30 sachets.

Product Descriptions:
Before reconstitution: Yellow colour powder with ginger flavour.
After reconstitution: Yellow colour solution with ginger flavour.

Pharmacodynamics:

- **Glucosamine** is a natural substance found in chitin, mucoproteins and mucopolysaccharides. It is involved in the manufacture of glycosaminoglycan, which forms cartilage tissue in the body; glucosamine is also present in tendons and ligaments. Glucosamine must be synthesised by the body but the ability to perform this decline with age. Glucosamine and its salts have therefore been advocated in the treatment of rheumatic disorders including osteoarthritis. Glucosamine also acts to improve the viscosity of synovial fluid by increasing synovial fluid production, thereby providing lubricant activity.

- **Chondroitin**, an acid mucopolysaccharide and important structural component of cartilage, is a constituent of most cartilaginous tissues. It is given orally in reactive arthritis such as gonococcal arthritis. Either taken together or individually, both chondroitin and glucosamine may provide chondroprotective action in bone, joint and connective tissue disorders. This combination therapy works synergistically to stimulate cartilage cell growth and provide strength and compression resistance to cartilage.

- **Methylsulfonylmethane (MSM)** is an organic sulphur-containing compound that occurs naturally in a variety of fruits, vegetables, grains and animals including humans. It blocks the inflammatory process and enhances the activity of cortisol, a natural anti-inflammatory hormone produced in the body. MSM is an effective natural analgesic and has been used as an application for pain relief over arthritis joints.

Pharmacokinetics:

- **Glucosamine**
Absorption: After oral administration, bioavailability of glucosamine is low due to first-pass hepatic metabolism~26%. The gastrointestinal absorption is close to 90%.
Distribution: Glucosamine is not protein-bound, but rather incorporates into plasma proteins (primarily globulins). Volume of distribution is 2.5 litres.
Metabolism: Liver, extensive as the first-pass effect in the liver in which more than 70% of glucosamine is metabolised.
Excretion: Renal excretion 10%; Faeces 11%.

- **Chondroitin**
Earlier studies using high molecular weight chondroitin concluded that there was no significant absorption of the high molecular weight version of chondroitin. More recent studies demonstrate that there is probably significant absorption of low molecular weight chondroitin. Absorption appears to occur from the stomach and small intestine. There is also an indication that some chondroitin, after absorption, does enter the joint space. In a study using depolymerised shark chondroitin, approximately 30% of the administered drug (as high and low molecular weight derivatives) was absorbed. Once absorbed, the chondroitin is incorporated into glycosaminoglycan-rich tissues such as joint cartilage.

- **Methylsulfonylmethane (MSM)**
Metabolism and excretion of MSM made the primary metabolite became detectable in serum approximately 2 hours after ingestion of MSM. With continued MSM ingestion, metabolite maintained a steady concentration in the serum. When MSM was stopped after 14 days, the mean metabolite concentration declined slowly over the subsequent 96 hours, and only trace amounts were detectable after 5 days. The decline in the serum metabolite was linear and its half-life appeared to be about 38 hours. Metabolite has been shown to persist in the blood up to 5 times longer than MSM.

Indications:
As an adjuvant therapy for osteoarthritis.

Dosage and Administration:
For oral administration only.
Adults: Take 1 sachet, once per day. Dissolve the powder in a glass of water (250ml) and drink immediately.
Children: Safety and effectiveness have not been established in children.

Symptoms and Treatments of Overdose:
No cases of accidental or intentional overdose are known. The animal acute and chronic toxicological studies indicate that toxic effects and symptoms of toxicity are not likely to occur, even after high overdose.

Contraindications:
It is contraindicated in patient with hypersensitivity to glucosamine, chondroitin or MSM.

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110mm (±2mm)

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260mm
(±2mm)

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As the active ingredient is obtained from seafood (shellfish), the product should not be given to patients who are allergic to shellfish.

Warning and Precautions:
This product treats the underlying cause of osteoarthritis and the therapeutic effect can only be seen after 2 to 3 weeks. Therefore, it is advisable to take an analgesic or anti-inflammatory drug if required during the first 2 to 3 weeks of therapy with this product.
Safety and effectiveness have not been established in children (< 18 years); therefore children (< 18 years) should avoid using this product.
The administration in patients with severe hepatic or renal insufficiency should be made under medical supervision.
A doctor should be consulted in order to exclude the presence of other joint conditions/diseases for which an alternative treatment should be considered.
This product contains active ingredients derived from seafood, therefore should not be given to patients who are allergic to shellfish.
This product contains chondroitin from bovine source.

Use in Pregnancy and Lactation:
Available evidence is inconclusive or inadequate for use in pregnant or lactating mothers. Until more information is available, this product should only be used under medical supervision in pregnancy and lactating mothers if the potential benefit to the mother justifies the potential risk to the foetus. Administration during the first 3 months of pregnancy must be avoided.

Interactions with Other Medicaments:
Effects on glucose metabolism and antidiabetic agents: It has been hypothesised that glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and/ or alteration of peripheral glucose uptake. Glucosamine may increase insulin resistance and consequently affect glucose tolerance. It may reduce antidiabetic agent effectiveness eg when used with these antidiabetic agents: Acarbose, Acetohexamide, Chlorpropamide, Glipizide, Glyburide, Metformin, Miglitol, Pioglitazone, Repaglinide, Rosiglitazone, Glimepiride, Tolbutamide and Troglitazone. Glucosamine is likely safe in patients with well-controlled diabetes (HbAlc less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose levels closely or more frequently.
Reduce effectiveness when used with glucosamine: Doxorubicin, Etoposide and Teniposide.
Warfarin: Elevations of International Normalized Ratio (INR) serum values and potentiation of anticoagulant effects. If concomitant therapy is necessary, the patient's INR should be more closely monitored.
Avoid combination with chitosan as it may form complexes with chondroitin sulphate, thus decreasing its absorption.

Undesirable Effects:
Cardiovascular: Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.
Central nervous system: Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%).
Gastrointestinal: Nausea, vomiting, diarrhoea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.
Skin: Skin reactions such as erythema and pruritus have been reported with therapeutic administration of glucosamine.

Storage Condition:
Store below 30°C. Protect from light and moisture.
Keep out of reach of children.
Jauhkan daripada capaian kanak-kanak.

Shelf Life:
3 years.

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Marketed By:
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A Subsidiary of YSPSAH

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